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Reason: Helium has the highest value of electron affinity among all the elements known

a) Both assertion and reason are true and reason is correct explanation for the assertion

b) Both assertion and reason are true but the reason is not the correct explanation for the assertion

c) Assertion is true and the reason is false

d) Both assertion and the reason are false

15. The electronic configuration of the atom having maximum difference in first and second ionisation energies is

a) $1s^2, 2s^2, 2p^6, 3s^1$ b) $1s^2, 2s^2, 2p^6, 3s^2$

c) $1s^2, 2s^2, 2p^6, 3s^2, 3s^2, 3p^6, 4s^1$ d) $1s^2, 2s^2, 2p^6, 3s^2, 3p^1$

16. Which of the following is second most electronegative element?

a) Chlorine b) Fluorine c) Oxygen d) Sulphur

17. IE_1 and IE_2 of Mg are 179 and 348 kcal mol^{-1} respectively. The energy required for the reaction $Mg \rightarrow Mg^{2+} + 2\ e^-$ is

a) +169 kcal mol^{-1} b) - 169 kcal mol^{-1}

c) + 527 kcal mol^{-1} d) - 527 kcal mol^{-1}

18. In a given shell the order of screening effect is

a) s > p > d > f b) s > p > f > d c) f > d > p > s d) f > p > s > d

19. Which of the following orders of ionic radii is correct?

a) $H^- > H^+ > H$ b) $Na^+ > F^- > O^{2-}$ c) $F > O^{2-} > Na^+$ d) None of these

20. The First ionisation potential of Na, Mg and Si are 496, 737 and 786 kJ mol^{-1} respectively. The ionisation potential of Al will be closer to

a) 760 kJ mol^{-1} b) 575 kJ mol^{-1} c) 801 kJ mol^{-1} d) 419 kJ mol^{-1}

21. Which one of the following is true about metallic character when we move from left to right in a period and top to bottom in a group?